Week 4 – Nested Loops & Strings

■ Learning Objectives

• Use nested loops to generate patterns and traverse 2D grids.
• Manipulate strings: indexing, slicing, and common methods.
• Combine loops with strings to solve CCC-style problems.

■ Key Concepts

1. Nested Loops: Outer loop controls rows; inner loop controls columns.
2. String Indexing: s[i]; Slicing: s[a:b:step].
3. Useful Methods: upper(), lower(), find(), replace(), strip(), split(), join().
4. Immutability: Strings can’t be changed in-place; build new strings.

Example 1 – Star Rectangle (Nested Loops)

```
rows = 3
cols = 5
for r in range(rows):
    line = ""
    for c in range(cols):
        line += "*"
    print(line)
```

Example 2 – Reverse and Slice Strings

```
s = "computing"
print(s[0], s[-1])       # first & last char
print(s[1:4])            # 'omp'
print(s[::-1])           # reverse
print(s.replace("ing", "er"))  # 'computer'
```

■ Practice Problems

Practice 1 – Hollow Box

Given width W and height H, print a hollow box of stars. Border is '*', inside is spaces.

Solution:

```
W = 8
H = 4
for r in range(H):
    if r == 0 or r == H - 1:
        print("*" * W)
    else:
        print("*" + " " * (W - 2) + "*")
```

Practice 2 – Count Vowels

Count the number of vowels in a string (case-insensitive).

Solution:

```
s = input().strip().lower()
vowels = "aeiou"
count = 0
for ch in s:
    if ch in vowels:
        count += 1
print(count)
```

Practice 3 – Grid Sum (String to Int)

Given lines like '1 2 3', read N lines and compute row sums (simulate 2D).

Solution:

```
n = int(input())
for _ in range(n):
    row = list(map(int, input().split()))
    print(sum(row))
```

■ Homework Challenge

Write a program to check if two words are anagrams (ignore spaces/case).

Expected Solution:

```
a = input().replace(" ", "").lower()
b = input().replace(" ", "").lower()
print(sorted(a) == sorted(b))
```